

Machine Learning–Based Virtual Sensor for Automotive Engine Combustion Monitoring

Keyword: combustion monitoring, virtual sensing, machine learning

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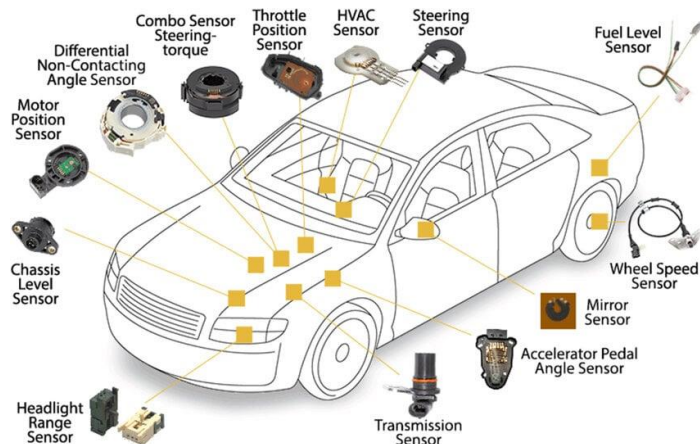
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Combustion Monitoring in Modern Engines

Critical combustion variables

- Intake air mass flow (MF_IA)
- NOx emissions
- Start of combustion (SOC)



Why matter?

- MF_IA
 - determines **cylinder air mass**
 - controls **oxygen availability for combustion**
- SOC
 - defines **ignition delay and heat release timing**
 - affects **thermal efficiency and pressure rise**
- NOx formation
 - governed by **temperature-dependent reaction kinetics**
 - strongly influenced by **oxygen concentration**



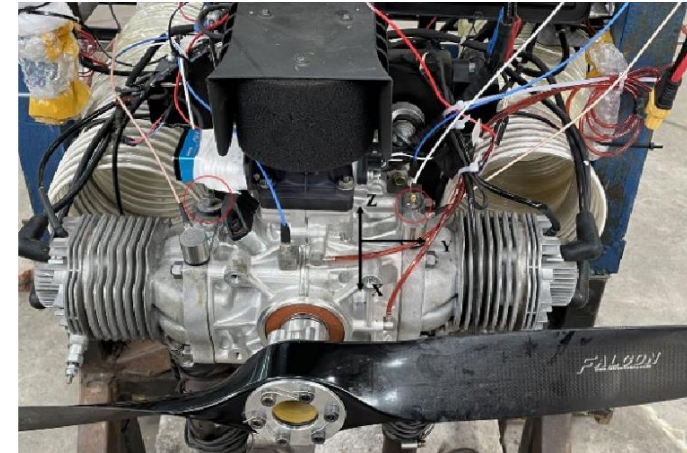
Limitations of Physical Sensors

Engineering limitations

- High hardware cost
- Packaging constraints in engine architecture
- Sensor drift and calibration requirements
- Long-term reliability and aging effects

System impact

- Additional sensors increase
 - system complexity
 - maintenance requirements
 - integration effort in ECU control loops



Source:
Small-sample Engine Fault Diagnosis Method Based on IACGAN and DNNs - Scientific Figure on ResearchGate.
Available from:
https://www.researchgate.net/figure/Failure-simulation-test-site-sensor-locations-are-marked-in-red_fig2_373189209 [accessed 13 Mar 2026]



Source:
Diagnosis of sensor faults in a combustion engine control system with the artificial neural network - Scientific Figure on ResearchGate. Available from:
https://www.researchgate.net/figure/Experimental-setup_fig3_334232843 [accessed 13 Mar 2026]



Virtual Sensor Concepts

Virtual Sensor Principle:

Model-based estimation of difficult-to-measure variables

Inputs

→ ECU signals representing **engine operating state**

Outputs

→ Combustion variables not directly measured

➤ **Physics perspective**

ECU signals indirectly encode

- air-handling dynamics
- fuel delivery and engine load
- thermodynamic boundary conditions

➤ **Key advantage**

Recover combustion information without additional hardware sensors

➤ **Research gap**

Most existing virtual sensors

→ estimate **one variable at a time**



Physical coupling between combustion variables

Airflow (MF_IA)

→ determines **oxygen availability**

EGR rate

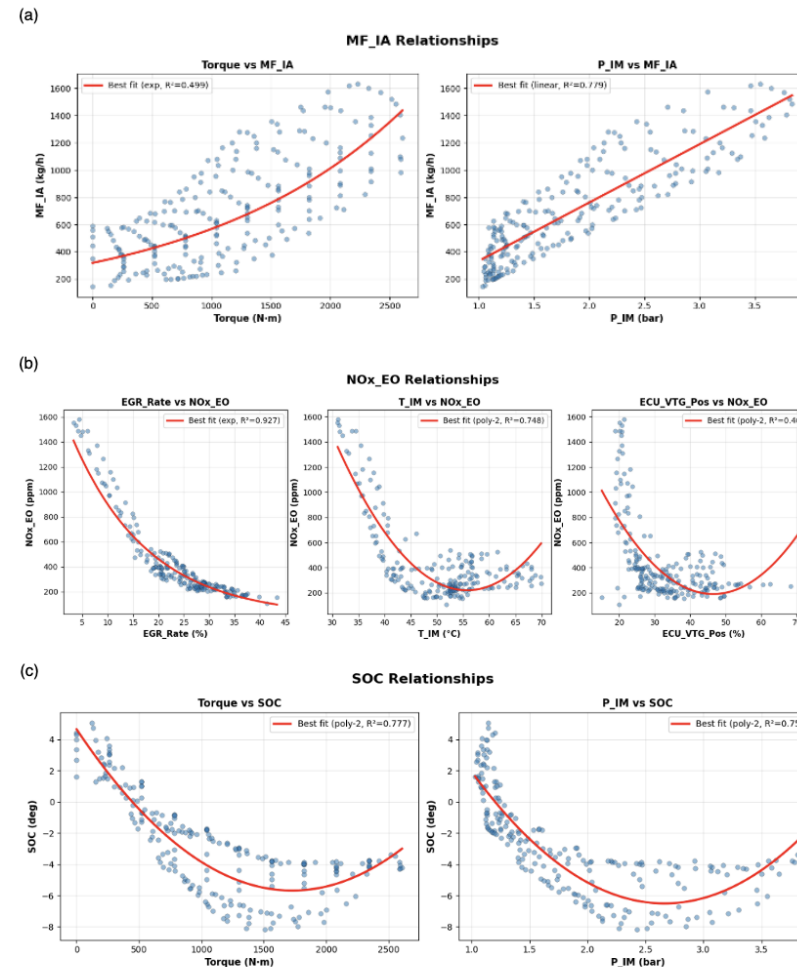
→ dilutes oxygen
→ reduces **combustion temperature**

SOC

→ controls **heat release phasing**

NOx formation

→ sensitive to **temperature and oxygen concentration**



Inter-correlation between outputs

- Airflow → influences SOC
- SOC → affects temperature field
- Temperature + oxygen → govern NOx formation



Research Objective

Develop a multi-target virtual sensor by using artificial neural network to simultaneously predict

- Intake airflow (MF_IA)
- Engine-out NOx emissions
- Start of combustion (SOC)

Key idea

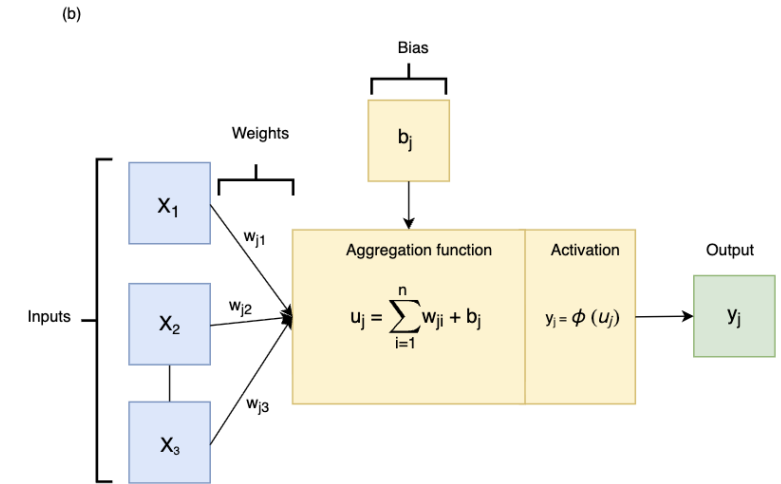
Shared-representation learning

Single model

- common internal representation
- multiple combustion outputs

Expected benefits

- exploit shared combustion physics
- improve predictive accuracy
- reduce redundancy among models



Virtual Sensor Model Framework

Experimental dataset

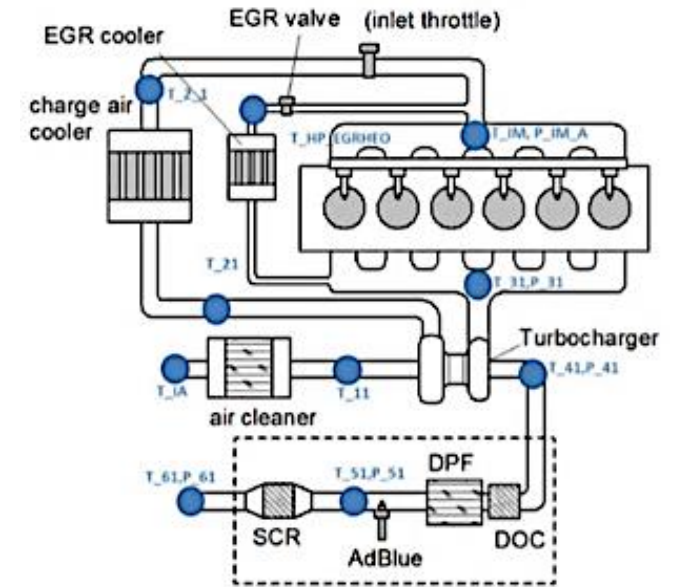
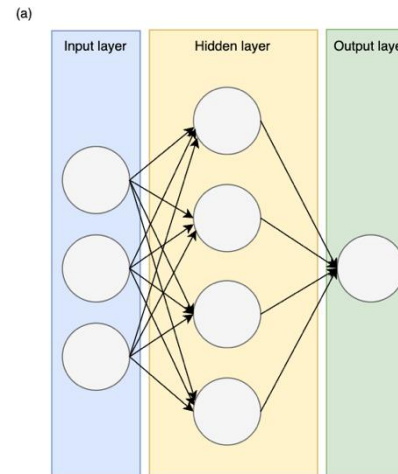
- 12-L six-cylinder heavy-duty diesel engine
- 217 steady-state operating points (80% training, 20% testing)
- Wide range of speed and load conditions

Model inputs (ECU signals)

- Engine torque
- Ambient pressure
- Intake manifold temperature
- Intake manifold pressure
- EGR rate
- Variable turbine geometry position

Outputs

- MF_IA
- Nox
- SOC



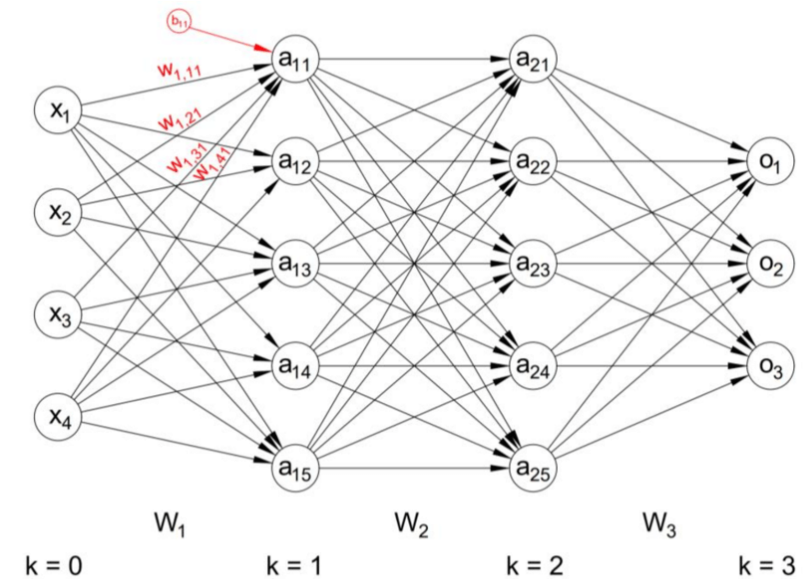
Benchmarking Methodology

Proposed model: Artificial Neural Network

- Structure: $6 \xrightarrow{W^{[3]}} 64 \xrightarrow{W^{[4]}} 32 \xrightarrow{W^{[5]}} 16 \xrightarrow{W^{[6]}}$
- nonlinear feature learning
- shared representation for multi-target prediction

Baseline models

- **OLS**
 - linear regression reference
- **Random Forest**
 - tree ensemble capturing nonlinear interactions
- **Support Vector Regression**
 - kernel-based nonlinear regression



Source: A Gentle Introduction to Physics-Informed Neural Networks, with Applications in Static Rod and Beam Problems - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/ANN-with-4-inputs-2-hidden-layers-with-5-neurons-and-3-outputs-Weights-of-connections_fig1_361148205 [accessed 9 Mar 2026]



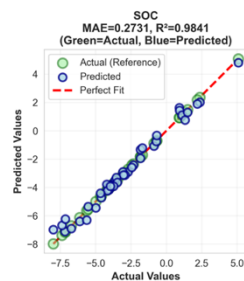
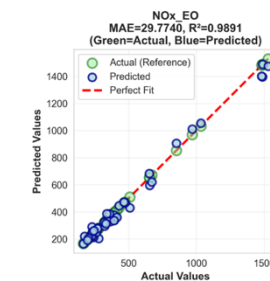
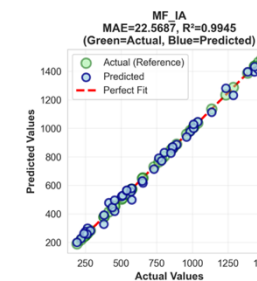


Results and Physical Interpretation

	Airflow (MF_IA)	Nox Formation	SOC Phasing
Dominant Inputs	- Intake pressure P_IM (ΔMAE 181.9) - Torque (ΔMAE 116.6)	- EGR rate (ΔMAE 203.3) - Intake temperature T_IM (ΔMAE 111.5)	- Torque (ΔMAE 1.878) - Intake pressure P_IM (ΔMAE 0.763)
Physical Meaning	- Intake pressure controls cylinder air mass - Engine load regulates air demand	- EGR reduces oxygen concentration - Temperature strongly influences thermal NOx	- Engine load affects ignition delay - Air supply influences combustion timing

Target	Metric	OLS	Random Forest	SVR (RBF)	ANN (6 inputs)
MF_IA	MAE	49.9	40.4	162.0	22.6
	R ²	0.973	0.981	0.647	0.995
NOx_EO	MAE	89.9	37.5	156.7	29.8
	R ²	0.878	0.974	0.244	0.989
SOC	MAE	1.30	0.40	0.22	0.27
	R ²	0.680	0.958	0.984	0.984

ANN Model: Predicted vs Actual - Clear Circle Overlap View (Test Set)



Limitations and Future Work

Limitation

- Training data
 - quasi-steady
 - moderate transients
- Possible degradation under
 - extreme transients
 - cold start
 - rare operating regimes
- Real-time deployment constraints
 - ECU computation limits
 - memory footprint
 - inference latency

Future Work

- Cross-engine transferability
- Hybrid physics + ML models
- Robustness to sensor drift and aging
- Validation under real driving cycles

